

BME 315 – Instrumentation laboratory (Spring 2025)

Results and discussion section of Lab Report 1

- Assessment SO3: An ability to communicate with a range of audiences.
- Performance criteria assessed:
 - a. Conveys technical information using graphics, tables, and charts.
 - b. Communicates technical information in written format.

Rubric used for the assessment of the results and discussion section of each lab report:

Section	Criteria Assessed	Points possible
Results and discussion	✓ All questions provided in the instruction have been answered correctly ✓ Proper use of figures, tables, captions, and titles. ✓ Data correctly discussed. ✓ Data included with the appropriate units. ✓ Analysis of data from the literature is discussed with references	20

Lab report 1 BME 315 – Results and discussion section only (instructions for students)

You should write the results and discussion section by:

1. Provide the standard curve of BSA. Include a figure caption below the graph. Do not include a title in the graph. You can use Excel to plot the graph. Make sure to label the axes with the units (5 points)
2. Calculate the concentration of the unknown protein concentration using your standard curve and report the value. Discuss whether the experimental data fits within the range of your standard curve. (5 points).
3. Point out any challenges in measuring proteins in your discussion. Discuss what are the key elements to ensure a proper reading when using a UV spectroscope (5 points)
4. Explain the study provided, discussing the study's objective and how this experiment differed from the one performed in the laboratory. Be sure to include the reference with Zotero when including the study in the laboratory report. (5 points)

Example of results and discussion lab report 1: Proficient (20/20)

Results and Discussion

The results of the experiment reveal the concentration of protein in an unknown sample through a standard curve.

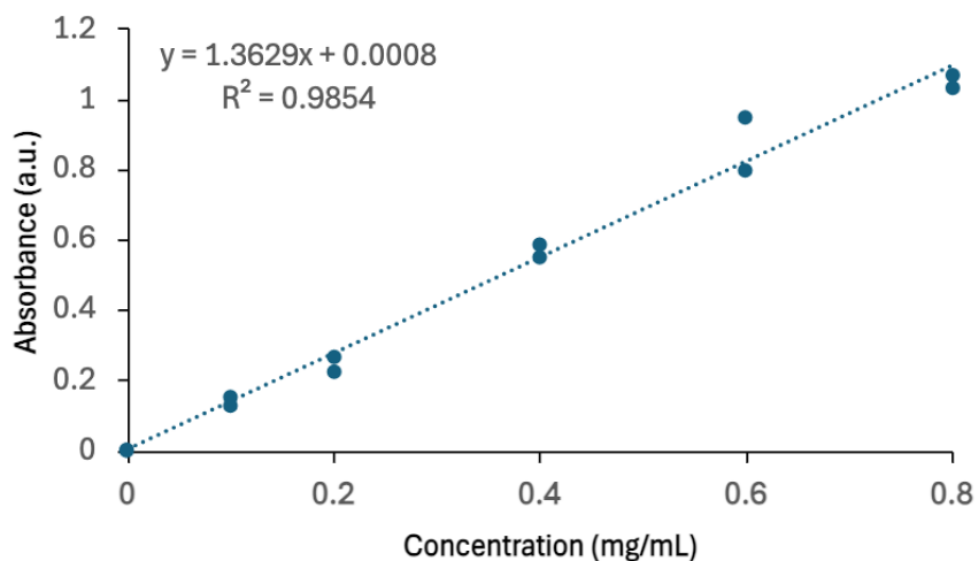


Figure 1. Standard curve of BSA dilutions in water at 595 nm. The curve is determined by a standard, BSA, at a known concentration undergoing dilution. The depicted linear range is valid for the Beer Lambert equation using two sets of data of absorbance measurements from 6 cuvettes containing the dilution samples. Results are reported as a mean $\pm$ standard deviation ($n=12$).

Absorbance of the unknown protein sample across three trials was measured to be 0.735, 0.732, and 0.773, respectively. Using the standard curve provided in Figure 1, the concentration of the unknown protein sample was calculated to be 0.547 ± 0.0168 mg/mL, of which fits in the range of 0.000 mg/mL to 0.800 mg/mL per the standard curve. A visual approximation of the experimental data against the standard curve also confirms the calculated concentration aligns with the trendline.

Employing UV spectroscopy as a method to measure protein concentration requires several calibration considerations for proper absorbance readings. Given the spectrometer is a piece of lab equipment that is electronically powered, it is necessary to ensure the spectrometer has been given ample time to warm up. This allows for the spectrometer to accurately measure

absorbance and provide relatively stable readings. Furthermore, as colorimetric assays draw on dyes that absorb at different maximum wavelengths, the spectrometer must be set at the relevant wavelength.

For a series of dilution measurements, measuring the absorbance of a blank cuvette must be done prior to measuring other samples. As the medium of the cuvette can absorb UV and visible light, it is pertinent to subtract the blank absorbance value for future sample readings. Similarly, the presence of bubbles in the cuvette can cause light scattering, in which light interacts with particles that deflect light from its intended path, rather than being absorbed by the sample, leading to inaccurate absorbance readings. Thus, all solutions were meticulously mixed to remove any bubbles. The procedural steps discussed were thoroughly carried out during the experiment to yield accurate absorbance readings. However, when conducting the experiment, there were instances in which the measured values of absorbance deviated from the accepted range of absorbance for a particular concentration (some cases of higher and lower than normal). Several samples were re-measured in order to obtain measurements that more closely aligned with the expected values. Such variance in absorbance readings was likely due to sample preparation and the sensitivity of the spectrometer. The last consideration is the generation of a linear range of absorbance readings in order to make use of the Beer-Lambert equation.

In a Bradford colorimetric assay experiment conducted by the Department of Chemical and Biological Engineering at Chalmers University of Technology, the viability of alternative drug delivery of therapeutic proteins was investigated. The study aimed to measure how protein quantification in a sample of a drug, by Bradford assay, was affected by the release of a polymer that is a component of the formulation of the drug¹⁰. In comparison to the experiment carried out in the laboratory, three polymers' absorbances (Carbopol, chitosan, and hydroxypropylmethylcellulose) were measured individually, in the acidic pH of the Bradford standard, and upon interaction with the Coomassie-blue dye. A model protein, ovalbumin (OVA) was then added into mixtures of protein, Coomassie-blue, and polymer, and thereafter analyzed how absorbance of OVA concentration at 595 nm was impacted by the addition of the polymers. It was found that two polymer additives, Carbopol and chitosan, affected the absorbance of the three-component mixture due to light scattering and dye interaction, particular occurrences considered in the lab. Beyond a standard curve of OVA, full spectra of absorption for the mixtures were generated from 350 nm to 850 nm in order to correct polymer interference as well as better analyze polymer effect on the Bradford assay. From there, corrective procedures were tested by subtracting absorption at 850 nm then scaling the result by a constant indicative of increasing polymer turbidity, or "cloudiness", at shorter wavelengths. The corrective equation was as follows¹⁰,

$$A_{corr}(595\text{ nm}) = A_{obsd}(595\text{ nm}) - KA_{obsd}(850\text{ nm}) \quad (2)$$

where K is equal to 1.1. Due to the polymer turbidity, this equation holds for Carbopol, rather than for chitosan.

Comment of the instructor: You have successfully conveyed the technical information of the report and correctly analyzed the data.

Example of results and discussion lab report 1: Meets (17/20)

Results and Discussion

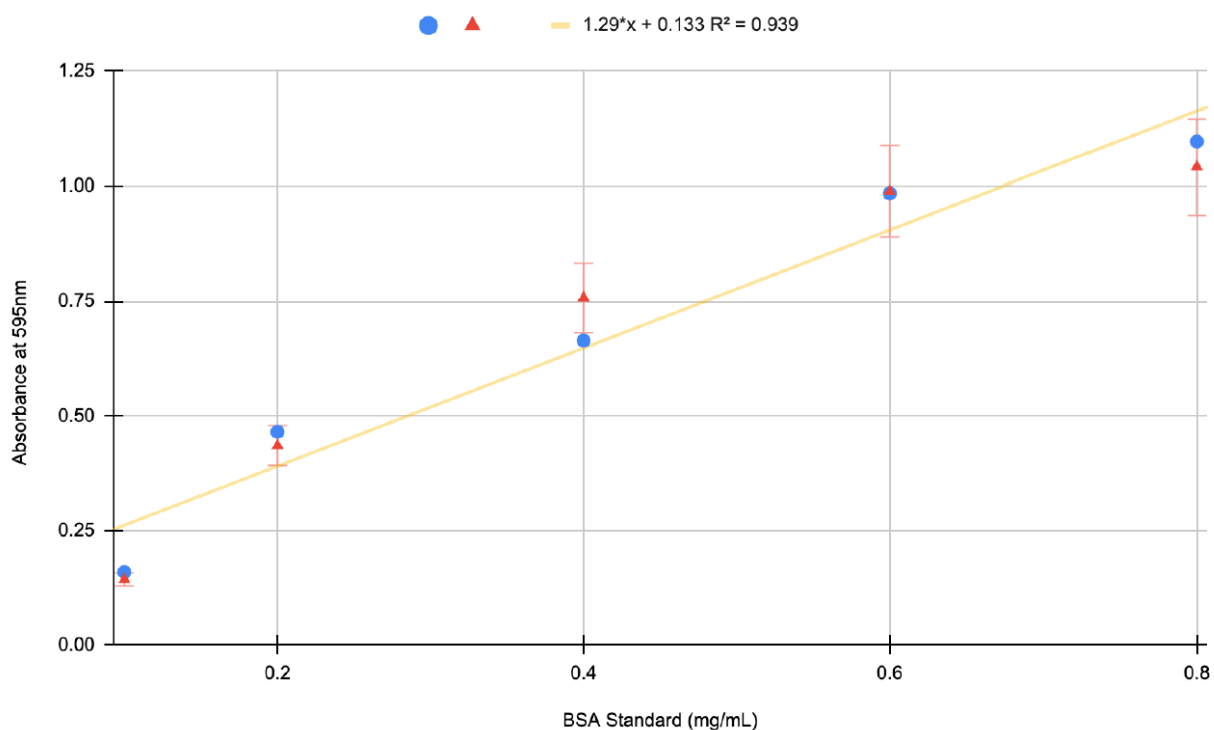


Fig.1. Absorbance at 595 nm of BSA samples versus protein concentration in measuring their absorbance values, testing trial 1 (red triangle) and testing trial 2 (blue circle) are in a closer range of the increasing curve as the BSA concentration increases in the dilution samples. Giving us a standard curve of $y = 1.29x + 0.133$ (the average between the two trials).

The concentration of the unknown protein concentration, given with the help of the graph, is $0.61 \pm 0.117 \text{ mg/mL}$ which fits in the standard curve of the graph. Calculating the standard deviation from the given unknown protein from the testing trials, the values don't all fit in the range of the standard curve. While testing the absorbance values of the unknown protein sample, its value differentiated more than the other results, giving the calculations a different range for the standard deviation value. This shows a potential error in the experimentation.

The measuring of proteins is complex, while analyzing them there were some challenges that had to be overcome. Some proteins just exist in low concentrations, which makes the detection equipment require a sensitive detection to be able to measure the low concentrations. Protein structure and composition are different in their characteristics and modifications; there is no universal method to be able to quantify them all the same.

Because of a protein's structure, other molecules that attach to it interfere with its reading. Another challenge that protein measurement faces is its method of measurement. With the different assay methods, each has its limitations in the way they show or in their process arrangements.

In this experiment, compared to the study provided, '*Quantification of protein concentration by the Bradford method in the presence of pharmaceutical polymers*,'⁶ there are several differences in the experiments. Although both experiments aim for the quantification of protein concentrations and the use of Bradford assay and absorbance method, they differ in multiple areas.

The experiment from the journal article tells us that they are investigating how pharmaceutical polymers such as Carbopol, Chitosan, and HPMC affect the Bradford assay. The journal article is about understanding the interaction with the Coomassie dye and its potential interference in protein concentration determination.⁶

Focusing on the effects of their experimental result on pharmaceutical polymers on the Bradford assay, their methodology differed from this lab. Their results and analysis recorded an absorption spectra of 350-850 nm, detecting polymer interferences showing that HMPC does not interfere with the Bradford method compared to the other polymers. This makes use of Ovalbumin (OVA), a pharmaceutical-relevant protein used for controlled drug release studies.⁷

In the experiment, the aim is to use the Bradford assay method for protein quantification. Finding standard curve, UV spectroscopy, and calculation of unknown protein concentrations. This experiment is done in lower levels compared to the pharmaceutical investigation, since learning the methods and how to use the tools and equipment needed is more relevant. This experiment is done in an undergraduate teaching lab with standard equipment and limited repetitions of the experiment.

The results of the experiment are shown with simple absorbance measurement at 595 nm for the protein quantification of the samples. The results and analysis of the experiment evaluated the accuracy of the Bradford assay base in the test trials made in the laboratory. The discussion was made around the fact that the factors affecting the protein measurement changed the results of the experimental values.

Comment of the instructor: Overall, the assignment meets the objective. However, there were a few issues in the discussion, as listed below.

Comment on the finding:

- When reporting the values of the unknown protein solution's concentration, ensure the standard deviation has the same number of significant figures as the average.
- You wrote: *Calculating the deviation from the given unknown protein from the testing trials, the values don't all fit in the range of the standard curve.*

This statement is not supported by the data. The standard deviation for the average protein concentration was ± 0.117 . All the values of absorbance obtained in the measurement were within the range.

Comment on the discussion:

- When discussing the challenges, you missed explaining why bubbles can interfere with absorbance readings. The main reason is the phenomenon of light scattering, so the actual absorbance would be higher than the actual reading because the light gets scattered and does not reach the detector. You could have also specified that below 220 nm, there is oxygen that interferes.
- When discussing the challenges, you could have mentioned that the solution needs to be homogeneously mixed to allow the color to be uniform across the cuvette. In addition, you could have mentioned that no scratches or visible dust should be present in the cuvette.
- You did a good job of identifying the key point of the paper. One point you should have pointed out is that the solvent used to solubilize the protein in our experiment was water, and no polymer was present in the solution.